Modelling data in Neo4j: Bidirectional relationships

October 11, 2013 · 3 min read

Neo4j

Transitioning from the relational world to the beautiful world of graphs requires a shift in thinking about data. Although graphs are often much more intuitive than tables, there are certain mistakes people tend to make when modelling their data as a graph for the first time. In this article, we look at one common source of confusion: bidirectional relationships.

Directed relationships

Relationships in Neo4j must have a type, giving the relationship a semantic meaning, and a direction. Frequently, the direction becomes part of the relationship's meaning. In other words, the relationship would be ambiguous without it. For example, the following graph shows that the Czech Republic defeated Sweden in ice hockey. Had the direction of the relationship been reversed, the Swedes would be much happier. With no direction at all, the relationship would

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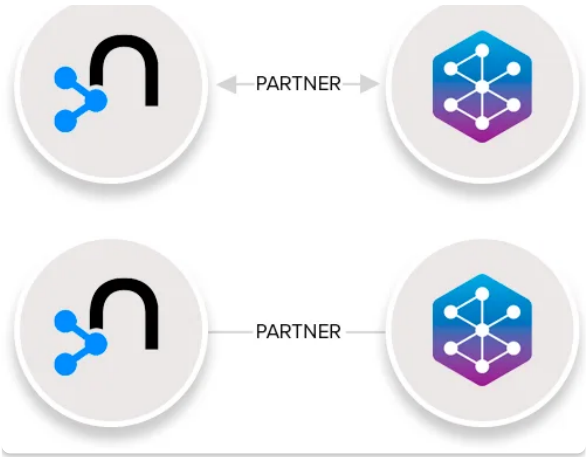
Note that the existence of this relationship implies a relationship of a different type going in the opposite direction, as the next graph illustrates. This is often the case. To give another example, the fact that Pulp Fiction was *DIRECTED_BY* Quentin Tarantino implies that Quentin Tarantino *IS_DIRECTOR_OF* Pulp Fiction. You could come up with a huge number of such relationship pairs.

One common mistake people often make when modelling their domain in Neo4j is creating both types of relationships. Since one relationship implies the other, this is wasteful, both in terms of space and traversal time. Neo4j can traverse relationships in both directions. More importantly, thanks to the way Neo4j organises its data, the **speed of traversal does not depend on the direction** of the relationships being traversed.

Bidirectional relationships

Some relationships, on the other hand, are naturally bidirectional. A classic example is Facebook or a real-life friendship. This relationship is mutual – when someone is your friend, you are (hopefully) his friend, too. Depending on how we look at the model, we could also say such a relationship is undirected.

GraphAware and Neo Technology are partner companies. Since this is a mutual relationship, we could



But since none of this is directly possible in Neo4j, beginners often resort to the following model, which suffers from the exact same problem as the incorrect ice hockey model: an extra unnecessary relationship.

Neo4j APIs allow developers to completely **ignore relationship direction** when querying the graph, if they so desire. For example, in Neo4j's own query language, Cypher, the key part of a query finding all partner companies of Neo Technology would look something like

```
MATCH (neo)-[:PARTNER]-(partner)
```

The result would be the same as executing and merging the results of the following two different queries:

`MATCH (neo)-[:PARTNER]->(partner)` and `MATCH (neo)<-[:PARTNER]-(partner)`

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Conclusion

Relationships in Neo4j can be traversed in both directions at the same speed. Moreover, direction can be completely ignored. Therefore, there is no need to create two different relationships between nodes if one implies the other.

Meet the authors

Michal Bachman

CEO

Neo4j certification

Michal Bachman, founder and CEO of GraphAware, drives the company's culture, core values, long-term vision, and strategy. With a strong technical background in computer science and engineering, he is passionate about using graph technologies, data science, and machine learning to solve the world's safety and security challenges. Michal

exploration to develop and manage GraphAware's key customer and partner relationships worldwide.

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